

# 664 HW3

Alex Caramanico

2025-04-21

## Lecture 7 Problems

1.

Suppose the slope estimates of Y on X from 5 imputed datasets are {1.5, 1.4, 1.6, 1.0, 1.3}; The variance estimates of slope estimates are {.21, .24, .30, .27, .25}. Using Rubin's combination formula, please calculate the combined estimate of the slope, its between-imputation variance, within-imputation variance, total variance, degrees of freedom, and the 95% confidence interval of the slope estimate.

```
slopes <- c(1.5,1.4,1.6,1,1.3)
variances <- c(0.21,0.24,0.3,0.27,0.25)
between_var <- sum((slopes - mean(slopes))^2) / (length(slopes)-1)
total_var <- mean(variances) + (1+1/length(slopes))*between_var
lambda <- (1+1/length(slopes))*between_var / total_var
rbind(Combined_Estimate = mean(slopes),
      Between_Var = between_var,
      Within_Var = mean(variances),
      Total_Var = total_var,
      DF = (length(slopes) - 1) / lambda^2,
      Lower95 = mean(slopes) + qnorm(0.025)*sqrt(total_var),
      Upper95 = mean(slopes) + qnorm(0.975)*sqrt(total_var))
```

```
##           [,1]
## Combined_Estimate 1.3600000
## Between_Var      0.0530000
## Within_Var       0.2540000
## Total_Var        0.3176000
## DF               99.7485859
## Lower95          0.2554425
## Upper95          2.4645575
```

```
rm(slopes,variances,between_var,total_var,lambda)
# that is a pretty wide confidence interval
```

## Lecture 8 Problems

1.

Using the real dataset that you used in previous lectures to multiply impute the missing variable. Perform multiple imputation analysis for the following estimands (a-c).

Compare the multiple imputation estimates with complete case analysis and discuss the results. Note that you need to decide if this missing variable is continuous or categorical first before deciding the imputation model. And choose several fully-observed variables as predictors. Also state clearly what regression analyses you want to run. (Similar to Examples 8.2 and 8.3)

```
library(mice)
library(tidyverse)
library(plotrix)
data("airquality")
complete_data <- na.omit(airquality)

# for simplicity, remove missing Solar.R values (so there are three fully
# observed variables) and time variables
airquality <- airquality[-which(airquality$Solar.R %>% is.na()),
                        c(1:4)]
# calculate proportion of missingness, set as M
M <- (sum(airquality$Ozone %>% is.na()) / nrow(airquality) * 100) %>% round()

imputations <- airquality %>%
  mice(method='norm',m=M,seed=827,print=F) # note to self - mice() can do a LOT
  #complete('long',include=T) # for actual data
```

a.

The mean of this missing variable

```
mean_imp <- imputations %>%
  with(exp=lm(Ozone~1)) %>%
  pool() %>%
  summary() # removes some other statistics but gives std error estimate
rbind(Multiple_Imputation=c(Mean=mean_imp$estimate,
                           StdErr=mean_imp$std.error),
      Complete_Case=c(Mean=mean(complete_data$Ozone),
                      StdErr=std.error(complete_data$Ozone)))
```

```
##              Mean   StdErr
## Multiple_Imputation 42.35998 2.809209
## Complete_Case      42.09910 3.158415
```

The results are extremely similar in this case, but the notable difference is that the standard error is much lower. However, this could simply be due to the imputed inherently having a larger sample size than the “complete” data.

b.

A regression analysis using this variable as the outcome variable and other variables (fully observed) in the dataset as the predictor

```

# This regression uses Solar Radiation, Wind, and Temperature as explanatory
# variables for Ozone, which has some missing values
reg1_imp <- imputations %>%
  with(exp=lm(Ozone~Solar.R + Wind + Temp)) %>%
  pool()
reg1 <-lm(Ozone~Solar.R+Wind+Temp,data=complete_data) %>% summary()
rbind(Multiple_Imputation=reg1_imp$pooled$estimate,
      Complete_Case=reg1$coefficients[,1])

```

```

##                (Intercept)      Solar.R      Wind      Temp
## Multiple_Imputation  -60.71732  0.06173740 -3.405568  1.608697
## Complete_Case        -64.34208  0.05982059 -3.333591  1.652093

```

The most notable difference is with the intercept, which is lower for the complete-case approach. The significance of each variable remains across all imputed data sets, suggesting complete case analysis *may* be sufficient for this data, although multiple imputation is more precise.

c.

A regression analysis that this missing variable is a predictor and one of the other fully observed variables is an outcome variable.

```

# This regression uses Ozone, Wind, and Solar.R as explanatory variables for
# Temperature, with Ozone having some missing values. Temperature was chosen as
# the response because it was been established to have a strong relationship
# with Ozone in Homework 1.
reg2_imp <- imputations %>%
  with(exp=lm(Temp~Ozone+Solar.R+Wind)) %>%
  pool()
reg2 <- lm(Temp~Ozone+Solar.R+Wind,data=complete_data) %>% summary()
rbind(Multiple_Imputation=reg2_imp$pooled$estimate,
      Complete_Case=reg2$coefficients[,1])

```

```

##                (Intercept)      Ozone      Solar.R      Wind
## Multiple_Imputation   72.04613  0.1694994  0.008227145 -0.2638417
## Complete_Case         72.41858  0.1719660  0.007275637 -0.3229446

```

```
rm(list=ls())
```

These results are also similar across methods - It can be said again that the relationship between the explanatory variables and the response is consistent across the implemented methods. However, they're "more" different in this model, especially the estimates for Solar.R and Wind. That is, the percentage difference between estimates across methods is greater when the missing variable is used as an explanatory variable rather than a response variable. However, it is not sufficient in this example for the significance or direction of the relationships to change, thus the interpretation of the results is similar. It could be argued that the only considerable difference would be in using the estimates to *predict* new values for temperature.

This result, along with the robustness of the results from parts a. and b., suggests that Ozone is probably MCAR. This is further supported by the t-tests conducted in Homework 1 (and illustrated by the box plots in said document). If the variable was missing at random (MAR), the last regression analysis (with the Ozone as the missing variable) likely would have resulted in very different results for multiple imputation and complete-case analysis.

## Lecture 9 Problems

1.

Continuing from the homework assignment from Lecture 8, using the real dataset that you used in previous lectures to multiply impute the missing variable. But this time try PMM method. And perform multiple imputation analysis for the following estimands. Compare the PMM imputation estimates with complete case analysis and previous imputation methods and discuss the results.

a.

The mean of this missing variable

```
data("airquality")
# three fully observed variables, remove time-series variables
airquality <- airquality[~which(airquality$Solar.R %>% is.na()),1:4]
# calculate proportion of missingness, set as M
M <- (sum(airquality$Ozone %>% is.na()) / nrow(airquality) * 100) %>% round()

complete_data <- na.omit(airquality) # for CC analysis

mean_impute_data <- airquality
mean_impute_data[which(mean_impute_data$Ozone %>% is.na()),1] <-
  mean(complete_data$Ozone) # for mean imputation analysis

# pmm imputations
imputations_1 <- mice(airquality, method='pmm', m = M, seed=8489,print=F)
# normal imputations
imputations_2 <- mice(airquality, method='norm', m = M, seed=8489,print=F)

mean_imp1 <- imputations_1 %>%
  with(exp=lm(Ozone~1)) %>%
  pool() %>%
  summary()

mean_imp2 <- imputations_2 %>%
  with(exp=lm(Ozone~1)) %>%
  pool() %>%
  summary()

# compare results
rbind(Predictive_Mean_Matching=c(Mean=mean_imp1$estimate,
                                StdErr=mean_imp1$std.error),
      Multiple_Imputation=c(Mean=mean_imp2$estimate,
                             StdErr=mean_imp2$std.error),
      Complete_Case=c(Mean=mean(complete_data$Ozone),
                       StdErr=std.error(complete_data$Ozone)),
      Mean_Imputation=c(Mean=mean(mean_impute_data$Ozone),
                         StdErr=std.error(mean_impute_data$Ozone)))
```

```
##                               Mean   StdErr
## Predictive_Mean_Matching 41.88470 2.930204
```

```
## Multiple_Imputation      42.66545  2.904918
## Complete_Case            42.09910  3.158415
## Mean_Imputation         42.09910  2.398649
```

```
rm(mean_imp1,mean_imp2)
```

Of course, the mean estimate is exactly same between mean imputation and the complete-case analysis, since the former simply imputes the latter's estimate for every missing value. This also has the effect of artificially reducing the variance, which is why mean imputation has the smallest standard error. It is interesting that the results are fairly different between PMM and normal multiple imputation, though the standard errors are nearly the same. PMM has the smallest estimate of the mean, while multiple imputation has the largest. The difference is likely due to slight differences in each algorithms' sampling procedure - PMM may favor smaller values in some way while MI favors larger. It could also be related to skewness in the Ozone variable.

b.

A regression analysis using this variable as the outcome variable and other variables (fully observed) in the dataset as the predictors

```
reg1_pmm <- imputations_1 %>%
  with(exp=lm(Ozone~Solar.R + Wind + Temp)) %>%
  pool()
reg1_mi <- imputations_2 %>%
  with(exp=lm(Ozone~Solar.R + Wind + Temp)) %>%
  pool()
reg1_cc <-lm(Ozone~Solar.R+Wind+Temp,data=complete_data) %>% summary()
reg1_mean <-lm(Ozone~Solar.R+Wind+Temp,data=mean_impute_data) %>% summary()
rbind(Predictive_Mean_Matching = reg1_pmm$pooled$estimate,
      Multiple_Imputation=reg1_mi$pooled$estimate,
      Complete_Case=reg1_cc$coefficients[,1],
      Mean_Imputation=reg1_mean$coefficients[,1])
```

```
##                (Intercept)      Solar.R      Wind      Temp
## Predictive_Mean_Matching  -68.49884  0.05851669 -3.168170  1.679493
## Multiple_Imputation      -65.39905  0.06314522 -3.303612  1.656134
## Complete_Case            -64.34208  0.05982059 -3.333591  1.652093
## Mean_Imputation          -45.53032  0.05459294 -2.780183  1.347863
```

```
rm(reg1_pmm,reg1_mi,reg1_cc,reg1_mean)
```

The mean imputation results are naturally quite far from the other results, due to the attenuation effects discussed in previous problems. That is, all of the parameter estimates are closer to zero, meaning the relationship between the predictors and the response is weakened due to mean imputation. The complete case and normal multiple imputation results are fairly comparable, but both are fairly different to the PMM results. / It is debatable whether this difference is considerable. It is unlikely that predicted values would vary too greatly across these parameter estimates, because the scale of the explanatory variables are not very large. However, some predictions may prove to be more accurate than others. More analysis would be incredibly useful here - perhaps setting aside a portion of the data to use for performance analysis (i.e. how far off each method is with its predictions). Another means of comparison would be model-fitting metrics, such as AIC or BIC. However, I am not sure what the combining methods would be for the multiple imputation approaches.

c.

A regression analysis that this missing variable is a predictor and one of the other fully observed variables is an outcome variable.

```
reg2_pmm <- imputations_1 %>%
  with(exp=lm(Temp~Ozone + Solar.R + Wind)) %>%
  pool()
reg2_mi <- imputations_2 %>%
  with(exp=lm(Temp~Ozone + Solar.R + Wind)) %>%
  pool()
reg2_cc <-lm(Temp~Ozone + Solar.R + Wind,data=complete_data) %>% summary()
reg2_mean <-lm(Temp~Ozone + Solar.R + Wind,data=mean_impute_data) %>% summary()
rbind(Predictive_Mean_Matching = reg2_pmm$pooled$estimate,
      Multiple_Imputation=reg2_mi$pooled$estimate,
      Complete_Case=reg2_cc$coefficients[,1],
      Mean_Imputation=reg2_mean$coefficients[,1])
```

```
##                (Intercept)      Ozone      Solar.R      Wind
## Predictive_Mean_Matching    72.70760 0.1648305 0.008925014 -0.3148743
## Multiple_Imputation         72.18975 0.1690897 0.007807766 -0.2744028
## Complete_Case               72.41858 0.1719660 0.007275637 -0.3229446
## Mean_Imputation             73.99243 0.1582994 0.011572407 -0.4690325
```

```
rm(reg2_pmm,reg2_mi,reg2_cc,reg2_mean,M,
   imputations_1,imputations_2,mean_impute_data)
```

Every result above is very similar, but naturally the mean imputation has the greatest “outliers” in its estimates. Interestingly, the parameter estimates are all *larger* in magnitude than the estimates for the other methods - the opposite of attenuation. This is due to swapping the missing variable to a predictor, rather than the response. Notably, PMM and normal multiple imputation are fairly similar for the intercept and Ozone (the missing variable), but not as much for Solar.R and Wind.

## 2.

Calculate the correlation between the missing variable and its response indicator with other variables that you used in the imputation model and present the results (see Example 9.3).

```
library(Hmisc) # for p-values
cor(complete_data) # base results
```

```
##           Ozone      Solar.R      Wind      Temp
## Ozone      1.0000000  0.3483417 -0.6124966  0.6985414
## Solar.R    0.3483417  1.0000000 -0.1271835  0.2940876
## Wind      -0.6124966 -0.1271835  1.0000000 -0.4971897
## Temp       0.6985414  0.2940876 -0.4971897  1.0000000
```

```
airquality$Observed <- !is.na(airquality$Ozone)
correlations <- rcorr(as.matrix(airquality))
rbind(Correlation_Observed=correlations$r[2:4,5],
      Pval=correlations$p[2:4,5])
```

```
##              Solar.R      Wind      Temp
## Correlation_Observed -0.02241613 -0.03248483 -0.06269504
## Pval                0.78826774  0.69709526  0.45218605
```

```
summary(glm(Observed ~ Solar.R, family='binomial', data=airquality))$coefficients
```

```
##           Estimate Std. Error   z value    Pr(>|z|)
## (Intercept)  1.264375315 0.453201686  2.7898734 0.005272866
## Solar.R      -0.000588742 0.002174185 -0.2707875 0.786554462
```

```
summary(glm(Observed ~ Wind, family='binomial', data=airquality))$coefficients
```

```
##           Estimate Std. Error   z value    Pr(>|z|)
## (Intercept)  1.37217271 0.59212589  2.3173665 0.02048377
## Wind        -0.02164297 0.05517032 -0.3922937 0.69484125
```

```
summary(glm(Observed ~ Temp, family='binomial', data=airquality))$coefficients
```

```
##           Estimate Std. Error   z value    Pr(>|z|)
## (Intercept)  2.4287352 1.70458059  1.4248286 0.1542067
## Temp        -0.0162418 0.02148275 -0.7560392 0.4496257
```

```
rm(airquality,complete_data,correlations)
```

The Observed of Ozone is not highly correlated with any of the other variables in the data set, and none are significant (even at  $\alpha = 0.1$ ). This is further backed up by the three logistic regressions, where none of Solar.R, Wind, or Temp are significant in explaining the variance in Observed. This is (once again) pretty strong evidence of the MCAR assumption with this data set, explaining why complete-case results are often very similar to the robust/proper missing data imputation methods.

## Lecture 10 Problems

### 1

For your real dataset, try to have more than one variable missing, that is, at least two variables have missing values. Use multivariate normal model to impute the missing data. And then come up with an analysis that includes these missing variables (e.g., a regression analysis) and compare the complete case analysis with multiple imputation results.

```
# Although airquality has some missing values for Solar.R, I chose a new data
# set that has higher proportions of missingness
library(norm)
data("mammalsleep")
# Columns: species, body weight(kg), brain weight(g), non-dream sleep (hrs/day),
#           dream sleep (hrs/day), total sleep (hrs/day), max lifespan (yrs),
#           gestation time (days), predation index (1=least),
#           sleep exposure index (1=least), overall danger index (1=least)
n_imputations <- (max(apply(mammalsleep,2, function(col) {is.na(col) %>% sum()}) /
                        nrow(mammalsleep))*100) %>% round()
```

```

### Complete-Case Analysis
complete_data <- na.omit(mammalsleep) # removes 20 rows/species
# total sleep omitted, since it's a linear combination of other sleep variables
cc_results <- lm(mls~bw+brw+sws+ps+gt+pi+sei+odi,data=complete_data) %>% summary()

### Multivariate Normal Imputation
variables <- names(mammalsleep)
mammalsleep <- as.matrix(mammalsleep[,-1])
s <- prelim.norm(mammalsleep)
thetahat <- em.norm(s,showits = F)
# getparam.norm(s,thetahat,corr=FALSE)
est <- std.error <- vector("list",n_imputations)
for (i in 1:n_imputations){
  rngseed(i) # set random seed
  newtheta <- da.norm(s, thetahat, steps=200) # 200 steps

  imp_data <- imp.norm(s, newtheta) %>% as.data.frame()

  names(imp_data) <- variables[-1] # species name not included

  # fit regression model
  lifespan_reg_norm <- lm(mls~bw+brw+sws+ps+gt+pi+sei+odi,data=imp_data)

  # save results
  est[[i]] <- summary(lifespan_reg_norm)$coef[, 'Estimate']
  std.error[[i]] <- summary(lifespan_reg_norm)$coef[, 'Std. Error']
}

# combine estimates
options(scipen=100,digits=6)
lifespan_reg_norm_mi <- mi.inference(est, std.error)

# compare results
cbind(Estimate=lifespan_reg_norm_mi$est,
      `Std. Error`=lifespan_reg_norm_mi$std.err,
      `t value` = lifespan_reg_norm_mi$est/lifespan_reg_norm_mi$std.err,
      `Pr(>|t|)`=lifespan_reg_norm_mi$signif,
      df=lifespan_reg_norm_mi$df)

```

|                | Estimate   | Std. Error | t value   | Pr(> t )      | df       |
|----------------|------------|------------|-----------|---------------|----------|
| ## (Intercept) | 27.9242101 | 9.33862395 | 2.990185  | 0.00280655997 | 3678.872 |
| ## bw          | -0.0237897 | 0.00676374 | -3.517242 | 0.00051546108 | 256.836  |
| ## brw         | 0.0266788  | 0.00578290 | 4.613390  | 0.00000430265 | 1483.814 |
| ## sws         | -0.3005618 | 0.62559030 | -0.480445 | 0.63105053957 | 752.892  |
| ## ps          | -2.7671840 | 1.74380733 | -1.586863 | 0.11391006567 | 231.236  |
| ## gt          | 0.0206599  | 0.02230350 | 0.926308  | 0.35431224458 | 8569.912 |
| ## pi          | -0.4123481 | 3.48092218 | -0.118459 | 0.90576272704 | 402.532  |
| ## sei         | 5.0374775  | 2.08317712 | 2.418170  | 0.01577397343 | 1014.899 |
| ## odi         | -6.6961612 | 4.85929094 | -1.378012 | 0.16898017195 | 394.758  |

```
cc_results$coefficients
```

```
##           Estimate Std. Error   t value    Pr(>|t|)
## (Intercept) 16.40366326 12.40327573  1.32252669 0.19508576700
## bw          -0.08082730  0.01727454 -4.67898460 0.00004735422
## brw          0.05343167  0.00942944  5.66647040 0.00000257836
## sws          0.00374556  0.67789565  0.00552528 0.99562475404
## ps          -1.87578772  2.03500242 -0.92176191 0.36334300090
## gt           0.02194337  0.02906487  0.75497915 0.45561624912
## pi          -2.60335104  4.52809865 -0.57493249 0.56923633767
## sei          5.63031605  2.46375761  2.28525567 0.02885626080
## odi         -2.03451628  6.42103805 -0.31685162 0.75335205937
```

```
# P-value differences
```

```
cbind(MVN_Imputation=lifespan_reg_norm_mi$signif,
      Complete_Case=cc_results$coefficients[,4],
      Difference=lifespan_reg_norm_mi$signif - cc_results$coefficients[,4])
```

```
##           MVN_Imputation Complete_Case   Difference
## (Intercept) 0.00280655997 0.19508576700 -0.19227920702
## bw          0.00051546108 0.00004735422  0.00046810685
## brw          0.00000430265 0.00000257836  0.00000172429
## sws          0.63105053957 0.99562475404 -0.36457421447
## ps          0.11391006567 0.36334300090 -0.24943293524
## gt          0.35431224458 0.45561624912 -0.10130400454
## pi          0.90576272704 0.56923633767  0.33652638937
## sei          0.01577397343 0.02885626080 -0.01308228736
## odi          0.16898017195 0.75335205937 -0.58437188742
```

```
# Correlation (collinearity) test
```

```
rcorr(mammalsleep %>% na.omit())$P %>% round(5)
```

```
##           bw      brw      sws      ps      ts      mls      gt      pi      sei
## bw          NA 0.00000 0.00990 0.63739 0.02624 0.00169 0.00000 0.54581 0.00770
## brw 0.00000      NA 0.01139 0.64015 0.02904 0.00001 0.00000 0.92299 0.03683
## sws 0.00990 0.01139      NA 0.00044 0.00000 0.01520 0.00002 0.02198 0.00006
## ps 0.63739 0.64015 0.00044      NA 0.00000 0.08575 0.00717 0.00905 0.00067
## ts 0.02624 0.02904 0.00000 0.00000      NA 0.01243 0.00001 0.00785 0.00001
## mls 0.00169 0.00001 0.01520 0.08575 0.01243      NA 0.00000 0.28254 0.04166
## gt 0.00000 0.00000 0.00002 0.00717 0.00001 0.00000      NA 0.56741 0.00007
## pi 0.54581 0.92299 0.02198 0.00905 0.00785 0.28254 0.56741      NA 0.00001
## sei 0.00770 0.03683 0.00006 0.00067 0.00001 0.04166 0.00007 0.00001      NA
## odi 0.09724 0.34001 0.00027 0.00008 0.00002 0.92645 0.04856 0.00000 0.00000
##           odi
## bw 0.09724
## brw 0.34001
## sws 0.00027
## ps 0.00008
## ts 0.00002
## mls 0.92645
## gt 0.04856
## pi 0.00000
```

```
## sei 0.00000
## odi      NA

rm(cc_results, complete_data, est, imp_data, lifespan_reg_norm, s, std.error,
   i, newtheta, thetahat)
```

My analysis idea was to see which variables are most strongly related to the life span of mammals. The documentation of the data set explains how it contains sleep, ecological, and constitutional variables. It is interesting to see which of these variable groups are most correlated with the animals' maximum life span. Several other analyses, such as how constitutional variables affect either type of sleeping, may also be worth exploring.

The results are actually fairly different between the multivariate normal imputation and the complete case analysis. The relationship direction only changed for one variable - sws ("nondreaming") sleep, as a predictor of maximum life span. However, it is nowhere near statistically significant by either approach. Perhaps the least surprising result is the extremely significant relationship between brain size (brw) and maximum life span. Typically, larger brains are associated with greater intelligence, which often in turn results in longer life spans. This relationship holds for either approach. The other significant variables in the complete case analysis are body weight (bw) and sleep exposure index (sei). Both of these are significant in the imputation approach as well. Interestingly, higher body weight seems to decrease life span, while sleep exposure seems to increase it (with negative and positive parameter estimates, respectively).

The third table shows the difference in p-values between the approaches, and the results are interesting as it shows that the multivariate imputation approach increases the p-value of some variables but decreases others - there is no "across the board" pattern. bw, brw, pi, and sei are brought further away from 0 (less significant) by multivariate imputation, shown by the positive difference in the table. On the other hand, the intercept, sws, ps, gt, sei, and odi are brought closer to 0 (more significant), shown by the negative difference. However, the only estimate that *becomes* statistically significant at  $\alpha = 0.05$  is the intercept, so it is unlikely that any conclusions would change. The "strangeness" of these results might be due to some collinearity between variables. An additional analytical idea might be to roll slow wave sleep and paradoxical sleep into one variable, perhaps as a ratio. This of course creates missing values in the case where either is not available, but could address any collinearity between the two.

The last table shows the correlation matrix of the complete data, revealing some collinearity concerns. There are *many* variables that have a statistically significant correlation with one another. This particular data set may benefit from the use of adjustment measures such as VIF to account for this potential issue. On the other hand, it might be argued that the three "index" variables should be included regardless of their correlations, because they each have distinct interpretations. The inclusion of particular variables is of course largely dependent upon subject matter expert opinion as well.

It is also worth noting that the data uses *maximum life span*, rather than the average or median. It is quite possible that some of the variable relationships would change if this metric was different, because intuitively a more-exposed animal surely has a lower average life span than a less-exposed one.

## Lecture 11 Problems

### 1

For your real dataset, try to have more than one variable missing, that is, at least two variables have missing values. Last time we used multivariate normal model to impute the missing data. This time try to use FCS imputation and compare with the results from the multivariate normal model imputation.

```
# normal models to impute any of the missing variables (shortwave sleep,
# paradoxical sleep, maximum life span, gestation time) seem reasonable. However,
# only the first variable follows some sort of "normal" curve, the others more
# closely resemble gamma distributions --> try PMM as well
```

```
fcs_imp1 <- mice(mammalsleep,
  method=c("", "", "norm", "norm", "", "norm", "norm", "", "", ""),
  maxit=5, m = n_imputations, seed=2405, print=F) %>%
  with(exp=lm(mls~bw+brw+sws+ps+gt+pi+sei+odi)) %>%
  pool() %>%
  summary()
```

```
fcs_imp2 <- mice(mammalsleep,
  method=c("", "", "pmm", "pmm", "", "pmm", "pmm", "", "", ""),
  maxit=5, m = n_imputations, seed=2405, print=F) %>%
  with(exp=lm(mls~bw+brw+sws+ps+gt+pi+sei+odi)) %>%
  pool() %>%
  summary()
```

```
# multivariate normal model imputation
```

```
data.frame(Estimate=lifespan_reg_norm_mi$est,
  `Std. Error`=lifespan_reg_norm_mi$std.err,
  `t value` = lifespan_reg_norm_mi$est/lifespan_reg_norm_mi$std.err,
  `Pr(>|t|)`=lifespan_reg_norm_mi$signif,
  df=lifespan_reg_norm_mi$df)
```

| ## |             | Estimate   | Std..Error | t.value   | Pr...t..      | df       |
|----|-------------|------------|------------|-----------|---------------|----------|
| ## | (Intercept) | 27.9242101 | 9.33862395 | 2.990185  | 0.00280655997 | 3678.872 |
| ## | bw          | -0.0237897 | 0.00676374 | -3.517242 | 0.00051546108 | 256.836  |
| ## | brw         | 0.0266788  | 0.00578290 | 4.613390  | 0.00000430265 | 1483.814 |
| ## | sws         | -0.3005618 | 0.62559030 | -0.480445 | 0.63105053957 | 752.892  |
| ## | ps          | -2.7671840 | 1.74380733 | -1.586863 | 0.11391006567 | 231.236  |
| ## | gt          | 0.0206599  | 0.02230350 | 0.926308  | 0.35431224458 | 8569.912 |
| ## | pi          | -0.4123481 | 3.48092218 | -0.118459 | 0.90576272704 | 402.532  |
| ## | sei         | 5.0374775  | 2.08317712 | 2.418170  | 0.01577397343 | 1014.899 |
| ## | odi         | -6.6961612 | 4.85929094 | -1.378012 | 0.16898017195 | 394.758  |

```
# normal FCS results
```

```
fcs_imp1
```

| ## |   | term        | estimate   | std.error   | statistic | df      | p.value      |
|----|---|-------------|------------|-------------|-----------|---------|--------------|
| ## | 1 | (Intercept) | 28.3368700 | 10.12273347 | 2.799330  | 40.0902 | 0.0078378697 |
| ## | 2 | bw          | -0.0243193 | 0.00507202  | -4.794802 | 44.7614 | 0.0000183366 |
| ## | 3 | brw         | 0.0261708  | 0.00578334  | 4.525214  | 45.0626 | 0.0000437693 |
| ## | 4 | sws         | -0.5010166 | 0.63124055  | -0.793701 | 40.2528 | 0.4320233962 |
| ## | 5 | ps          | -2.4692432 | 1.50452660  | -1.641209 | 37.9443 | 0.1090183311 |
| ## | 6 | gt          | 0.0234981  | 0.02647140  | 0.887678  | 44.7632 | 0.3794591222 |
| ## | 7 | pi          | -0.4946858 | 4.28814051  | -0.115361 | 27.8027 | 0.9089878562 |
| ## | 8 | sei         | 5.0759339  | 2.25669511  | 2.249278  | 39.6518 | 0.0301187467 |
| ## | 9 | odi         | -6.2906253 | 5.76888574  | -1.090440 | 26.7318 | 0.2852513089 |

```
# PMM FCS results
```

```
fcs_imp2
```

```
##           term      estimate  std.error  statistic      df      p.value
## 1 (Intercept) 27.8468234  9.45898085   2.9439560  43.5562  0.00518025918
## 2           bw -0.0251016  0.00488553  -5.1379478  46.0774  0.00000550447
## 3          brw  0.0270449  0.00565864   4.7793987  45.8612  0.00001845673
## 4          sws -0.3078837  0.61708390  -0.4989332  43.7622  0.62032527278
## 5           ps -3.0785014  1.48642372  -2.0710793  45.4998  0.04405115652
## 6           gt  0.0235009  0.02611862   0.8997761  45.4228  0.37298501707
## 7           pi  0.2073537  3.87727131   0.0534793  34.2076  0.95766114507
## 8          sei  4.9660952  2.10407380   2.3602286  44.5292  0.02270693741
## 9          odi -7.0037106  5.06485947  -1.3828045  35.5475  0.17535036195
```

```
# estimates summary table
```

```
cbind(Reg_Norm=lifespan_reg_norm_mi$est,
      Norm_FCS=fcs_imp1[,2],
      PMM_FCS=fcs_imp2[,2])
```

```
##           Reg_Norm  Norm_FCS  PMM_FCS
## (Intercept) 27.9242101 28.3368700 27.8468234
## bw          -0.0237897 -0.0243193 -0.0251016
## brw          0.0266788  0.0261708  0.0270449
## sws          -0.3005618 -0.5010166 -0.3078837
## ps          -2.7671840 -2.4692432 -3.0785014
## gt           0.0206599  0.0234981  0.0235009
## pi          -0.4123481 -0.4946858  0.2073537
## sei          5.0374775  5.0759339  4.9660952
## odi         -6.6961612 -6.2906253 -7.0037106
```

```
# variance / std error summary table
```

```
cbind(Reg_Norm_EstimateErr=lifespan_reg_norm_mi$std.err,
      Norm_FCS_EstimateErr=fcs_imp1[,3],
      PMM_FCS_EstimateErr=fcs_imp2[,3])
```

```
##           Reg_Norm_EstimateErr Norm_FCS_EstimateErr PMM_FCS_EstimateErr
## (Intercept)          9.33862395          10.12273347          9.45898085
## bw                   0.00676374           0.00507202           0.00488553
## brw                   0.00578290           0.00578334           0.00565864
## sws                   0.62559030           0.63124055           0.61708390
## ps                   1.74380733           1.50452660           1.48642372
## gt                   0.02230350           0.02647140           0.02611862
## pi                   3.48092218           4.28814051           3.87727131
## sei                   2.08317712           2.25669511           2.10407380
## odi                   4.85929094           5.76888574           5.06485947
```

```
# significance summary table
```

```
cbind(Reg_Norm_Pval=lifespan_reg_norm_mi$signif,
      Norm_FCS_Pval=fcs_imp1[,6],
      PMM_FCS_Pval=fcs_imp2[,6])
```

```
##           Reg_Norm_Pval Norm_FCS_Pval  PMM_FCS_Pval
## (Intercept) 0.00280655997  0.0078378697  0.00518025918
## bw          0.00051546108  0.0000183366  0.00000550447
## brw          0.00000430265  0.0000437693  0.00001845673
```

```
## sws      0.63105053957  0.4320233962  0.62032527278
## ps       0.11391006567  0.1090183311  0.04405115652
## gt       0.35431224458  0.3794591222  0.37298501707
## pi       0.90576272704  0.9089878562  0.95766114507
## sei      0.01577397343  0.0301187467  0.02270693741
## odi      0.16898017195  0.2852513089  0.17535036195
```

```
rm(list=ls())
```

The first table shows the original results from multivariate normal imputation, while the second and third show the fully-normally-specified FCS approach and the PMM-specified FCS approach, respectively. The fourth and fifth show only the parameter estimates and corresponding standard errors, for convenience. The last table shows the p-values for each parameter, so comparisons can be made on the significance of variables toward maximum life expectancy across each approach.

Naturally, with the first model specification, the results are extremely similar between the two methods. This is because normal models are used in the FCS approach, which effectively does the same thing as the multivariate normal imputation. There are only minor differences in the parameter estimates, due to the difference in random seed(s).

The second model specification (using PMM) is fairly different from the other two, with the estimate for predation index changing signs. Overall danger index and paradoxical sleep had larger parameter estimates using the PMM method for the missing variables, while all the remaining had weakened relationships (or were extremely similar, in the case of gestation time, birth weight, and brain weight). It is perhaps most worth discussing the differences in parameter estimates for variables that had missing values - 1) shortwave sleep, 2) paradoxical sleep, and 3) gestation time. For 1), PMM\_FCS and multivariate normal imputation are extremely similar, while Normal\_FCS is fairly different. For 2), each estimate differs by 0.3, with PMM\_FCS having the strongest negative estimate, Normal\_FCS the weakest. 3), on the other hand, is extremely similar for all three variables.

Comparing standard errors and p-values, there are some changes between the multivariate approach and Normal\_FCS, again due to differing random number generation. However, the significance conclusion does not change for any of the variables. Only birth weight, brain weight, and sleep exposure index are significant in explaining the variance in maximum life expectancy. However, *paradoxical sleep is significant under the PMM approach*. This could be very note-worthy for researchers, and the difference is eye-opening. The intercept is significant across all approaches.

Other data sets (with different types of variables) would likely benefit from exploring numerous modeling options, such as those presented in the example code.